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Product Development – Practice Overview 2024

Russian version of this page.

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> 1. This course accepts work from either groups of two or individuals.
> 2. JIRA could be replaced by any other alternative technology.

Hey there! This course is an attempt to think about and try to understand the ins and outs of product development. You'll go from brainstorming an idea to launching a fully-fledged app, all while learning the ropes of project management, design, and coding. Each step builds on the last, giving you a hands-on experience that's as close to the real deal as it gets. By the end of this course, you'll have a product under your belt and a toolbox full of skills that'll make you a real contender in the product development world.

> 💡 The teacher assistant is your go-to for setting the pace and the playbook—they'll define the ***report formats and deadlines***. You'll be tasked with crafting a full-fledged project, with a keen eye on design and planning. This isn't about winging it; every decision you make needs to be backed up with ***solid reasoning***.

Lab 1: Picking Your Project Idea

First up, you've got to pick a project idea. No pressure, but you gotta lock this down by **September 20th.** You'll need to do some serious digging into what's hot, who you're up against, and who your users are. Draw up some diagrams to map out how users will interact with your product. List out all the features you think you'll need and throw in some rough estimates.

> Remember, the teacher assistant is playing the customer here, so they might throw in some curveball feature requests during your defense. Make sure you've got a solid argument for every decision you make.

Acceptance criteria:

1. **Project Topic Selection** - Finalize your project topic by September 20th, with the option for the teacher assistant to define it.
2. **Business Analysis (BA)** - Conduct a comprehensive analysis of industry trends, competitors, and user audits.
3. **USE-CASE UML Diagrams** - Develop diagram(s) to illustrate user interactions with the product.
4. **Feature List with Estimation** - Compile a detailed list of required features, including time and resource estimations.

Lab 2: Plan & Design

You'll need to break down your feature list into bite-sized tasks and document everything clearly. Set up your Jira backlog with epics, labels, and estimation points to keep track of what's coming up. Draw up a Gantt diagram to keep an eye on your timeline. Also, provide a documented outline your Git workflow and tech stack. This is where you lay the groundwork for smooth sailing through development.

Acceptance criteria:

1. **Requirement Decomposition Documentation** - Systematically break down and document the project requirements for clarity and manageability.
2. **Jira Ticket Backlog Setup** - Create a structured backlog in Jira, categorizing tasks into epics, applying labels, and assigning estimation points, documented ticket format (included acceptance criteria).
3. **Gantt Diagram Creation** - Develop a Gantt diagram to visualize the project schedule, highlighting key milestones and dependencies.
4. **Git Workflow Definition** - Establish a clear Git workflow to streamline version control and facilitate team collaboration, merge strategies, commit message formats(**documented**).
5. **Technology Stack Specification** - Detail the selection of technologies, tools, and frameworks that will be used throughout the project.

Lab 3: Rolling Up Your Sleeves for Development

Time to code! You'll start by whipping up an MVP—it doesn't have to be perfect, just a solid base to build on. Document your system API and data formats so everyone's on the same page. Throw in some component diagrams and system design schemas to show how everything fits together.

Acceptance criteria:

1. **Minimum Viable Product (MVP) Development** - Build an initial version of the application, acknowledging that it may not include all planned features.
2. **System API and Data Formats Documentation** - Document the system's API and the data formats used for communication.
3. **Test-case Creation** - Develop a comprehensive set of test cases (TDD would be a plus).
4. **Component Diagram (UML)** - Create a UML component diagram to illustrate the relationships and interactions between system components (*main app*).
5. **System Design Schema** - Design a schema that outlines the architecture and structure of the system, ensuring scalability and maintainability.

Lab 4: Launch time!

You're almost there! First, draft a user guide to help folks get the hang of your app. **Deploy** the release version so it's live and kicking. Then, compile a report with all the stats from JIRA and GIT, plus a rundown of any issues you faced along the way and your release documentation. The end lab will require you a presentation to show off your hard work.

Acceptance criteria:

1. **User Guide Compilation** - Prepare a detailed user guide to assist end-users in understanding and effectively using the application.

2. **Deployed Release Version** - Ensure the final version of the application is fully deployed and accessible to users.
3. **Comprehensive Report** - Assemble a report that includes JIRA and GIT statistics, a candid assessment of challenges faced, and comprehensive release documentation.
4. **Project Presentation** - Develop and deliver a presentation to showcase the project's development, outcomes, and learnings.

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